Rational mNAV

Valuing the Wrapper in Bitcoin Terms Against the Self-Custody Hurdle

A Bitcoin-Denominated Monte Carlo Framework for Bitcoin-Treasury Equity: Strategy (MSTR) versus Self-Custody

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June 29, 2026

First published June 26, 2026

The author welcomes feedback on ways to improve this document and revises it as new learnings emerge and strategies evolve.

Abstract

We develop a valuation framework for a Bitcoin-treasury company—one whose enterprise is the accumulation of Bitcoin (BTC) funded by issuing equity, preferred stock, and convertible debt—taking the perspective of an investor whose unit of account is Bitcoin itself. The central question is not whether the equity rises in dollars but whether owning it delivers *more Bitcoin per dollar committed* than simply holding self-custodied coins. We answer it with a Monte Carlo engine that (i) simulates the BTC price under a *selectable* process (a power-law trend with mean-reverting residual, a geometric Brownian motion, or a diminishing-returns trend), each anchored to its own estimate of *today's fair value*—the power law to its formula, the other two to a lag-corrected 200-week moving average—and optionally overlaid with an auto-calibrated stochastic-volatility process, (ii) simulates the short policy rate (SOFR) under a selectable process (Vasicek, Cox–Ingersoll–Ross, random walk, or jump-diffusion) with an optional fiscal-dominance cap on the policy rate and optional joint fiscal-stress shocks that re-rate Bitcoin in the same event, (iii) propagates an explicit capital-structure engine in which the floating preferred (STRC) dividend is set endogenously off the simulated short rate (SOFR), the firm's leverage, the USD reserve runway, and a migrating credit rating, the STRC price is then *derived* as the ratio of that coupon to the yield the market requires—so its discount to par, and hence the fraction of the time its at-the-market program can issue at or above par, are outputs of the Bitcoin and reserve path rather than assumptions; convertible notes resolve contingently at their put dates under an assumed-diluted share count, a proactive accretive equity program issues into strength at a pace that scales with the premium, and residual financing needs are met through a strict waterfall (free cash flow, cash reserve, at-the-market equity, at-the-market preferred, and—only as a last resort—BTC sales), and (iv) prices each share at exit as its gross Bitcoin backing per share multiplied by the premium the market grants—a *structural* premium (a net-backing ratio of Bitcoin value less the senior claims ranking ahead of the common, an option/convexity component, and a convex upside "flywheel" that lets a volatility spike balloon the premium when the firm is strong) evaluated under *the market's* Bitcoin outlook and recorded as the median premium path the price reverts to and follows, while *the investor's* own outlook drives the Bitcoin price and hence the backing per share. Three finite-supply brakes—execution slippage on the firm's own buys, a concentration-driven decay of the premium, and a preferred-carry test—cause the terminal Bitcoin holding to emerge as a derived asymptote rather than an input. The investor's rule is to prefer the equity only when its expected *Bitcoin-denominated* return clears a counterparty/wrapper-risk hurdle relative to self-custody. From this we derive a closed

expression for fair value—the share price at which the median Bitcoin-denominated return equals the hurdle, equivalently the price at which the probability of beating self-custody is one-half—and we translate it into a *rational mNAV*. We distinguish the *market's* rational mNAV—the premium the price commands, set by the market's outlook—from the investor's *personal* rational mNAV, the fair entry under the investor's own outlook and hurdle. The rational premium is thus not a single number but a function of beliefs. All figures are conditional outputs of the stated assumptions and are not forecasts.

1 Motivation and numeraire

A Bitcoin-treasury company presents a deceptively simple question. If one wishes to own Bitcoin, is it better to buy the coins and hold the keys, or to buy a share of a company that *owns* coins on its own balance sheet and finances additional purchases with leverage? A treasury company is not a custodian: it owns the Bitcoin outright, and the shareholder owns equity whose value derives from those coins but cannot be redeemed for them. There is no right of withdrawal—one cannot exchange shares back into the underlying coin—so the shareholder's claim is mediated entirely by the market price of the stock and by the corporate structure sitting between the coins and the equity. The two routes are therefore not equivalent: the equity wrapper layers on a premium or discount to net asset value, a senior preferred and convertible stack, dilution at management's discretion, and assorted counterparty, governance, and regulatory risks; in exchange it offers leveraged accumulation that, when it works, grows the Bitcoin backing *per share* faster than an individual could on their own.

The cleanest way to adjudicate is to change the unit of account. We denominate everything in Bitcoin. Holding one's own coin is then the risk-free benchmark: one Bitcoin remains one Bitcoin, a Bitcoin-denominated return of zero. The equity is worth buying only if its Bitcoin-denominated return is positive after a hurdle that compensates for the wrapper's incremental risk. This reframing removes the dollar price of Bitcoin as a confound (it cancels from both sides) and isolates the only thing that matters: whether the corporate structure is expected to deliver more coin per share than one started with.

2 The decision rule

Let B_t be the USD BTC price, H_t the coins held by the firm, N_t the diluted share count, and $b_t \equiv H_t/N_t$ the *Bitcoin backing per share*. Let m_t denote the market premium ("mNAV"), defined throughout this paper as the ratio of equity market capitalization to the value of the firm's Bitcoin,

$$m_t \;=\; \frac{N_t P_t}{H_t B_t}, \tag{1}$$

where P_t is the share price. An investor who buys one share at $t = 0$ pays, in Bitcoin terms, $P_0/B_0 = m_0\, b_0$ coins. At horizon τ the share is worth $b_T\, m_T$ coins. The Bitcoin-denominated gross multiple of the position is therefore

$$G \;=\; \frac{b_T\, m_T}{m_0\, b_0}, \tag{2}$$

against a benchmark multiple of 1 for self-custody. Writing r_p for the annual counterparty/wrapper-risk hurdle, the rule is to prefer the equity whenever

$$G^{1/\tau} - 1 \;>\; r_p. \tag{3}$$

Equivalently, the hurdle compounds over the full holding period: the equity is preferred whenever its Bitcoin-denominated exit value exceeds the entry cost grown at r_p for τ years,

$$b_T\, m_T \;>\; m_0\, b_0\, (1 + r_p)^{\tau}. \tag{4}$$

The hurdle is therefore an *annual* rate, not a one-time threshold: a 3.5% hurdle over a ten-year hold demands a Bitcoin-denominated gain of $(1.035)^{10} - 1 \approx 41\%$ across the horizon, not 3.5%.

Equation (2) cleanly separates the two engines of value: the growth in backing per share b_T/b_0 (the accretion "flywheel"), and the change in the premium m_T/m_0 (the market's willingness to keep paying above net asset value). As Section 5.2 makes precise, the premium term is governed by *the market's* Bitcoin outlook rather than the investor's, while the investor's own outlook drives the Bitcoin price beneath b_T; everything that follows is machinery for simulating the joint distribution of b_T and m_T.

The sign of the hurdle. A positive r_p is the usual case: the wrapper carries incremental counterparty, governance, dilution, and regulatory risk, and a rational holder demands compensation for bearing it. But r_p is a personal parameter, and for some investors it is legitimately *negative.* Consider someone who wants Bitcoin exposure yet cannot or will not self-custody—wary of their own key management, bound by institutional or fiduciary constraints, or simply unwilling to be their own bank—and who also wishes to avoid the recurring management fee of a spot ETF, a drag that compounds against the holder every year the position is held. For that investor the equity delivers Bitcoin exposure with the prospect of a *growing* Bitcoin-denominated balance (the accretion flywheel), no annual fee, and a custodian they trust more than themselves. The convenience, the avoided ETF fee, and the trust carry real value—the mirror image of the recurring fee a spot ETF charges—so such an investor demands no positive premium over self-custody to hold the wrapper, and may rationally assign it a *negative* hurdle. That lowers the bar in (3) and, through the fair-value relation (23), *raises* the rational mNAV—the wrapper can be worth a premium to its own Bitcoin backing for someone who values what it provides beyond the coins themselves.

2.1 Why mNAV should be common-equity market cap over total Bitcoin holdings

The numerator and denominator of the premium are a modeling choice, and three conventions are in common use among analysts. They are not interchangeable: each measures a distinct quantity and answers a distinct question. We state the three explicitly and justify our selection on analytical rather than presentational grounds.

Let P denote the MSTR common share price, N the assumed-diluted common share count, H the quantity of Bitcoin held, B the Bitcoin spot price, D the outstanding convertible principal, and L the aggregate preferred liquidation preference. The three measures are

$$m^{\text{eq}} = \frac{PN}{HB}, \qquad m^{\text{EV}} = \frac{PN + D + L}{HB}, \qquad m^{\text{net}} = \frac{PN}{HB - D - L}. \tag{5}$$

We adopt the first, m^{eq}—the market capitalization of the common equity over the gross value of the Bitcoin holdings.

A terminological caveat is in order. Strictly, "mNAV" is a misnomer for the quantity we adopt. A *net* asset value would subtract the firm's liabilities and add its non-Bitcoin assets—here, principally the USD reserve—and m^{eq} does neither. It is simply the ratio of the (assumed fully diluted) common-equity market capitalization to the gross market value of the Bitcoin holdings, with nothing netted out and no other assets included. A more accurate label would be a *multiple of asset value*—“mAV”—since the denominator is a gross asset value rather than a net one. We nonetheless retain “mNAV” throughout this paper, because the term is in common and well-understood use in discussions of Bitcoin-treasury equities; coining a private acronym would cost more in legibility than the added precision would return. The reader should simply keep in mind that the “N” is, in our usage, a convention rather than a literal description.

The justification is that the decision this framework addresses is posed from the standpoint of the residual (common) claimant, whose objective is the Bitcoin-denominated total return on the common over a multi-year horizon. The governing state variable is the Bitcoin backing per common share, H/N, and its evolution under the financing program. Of the three ratios, m^{eq} is the only one whose denominator is an observable, assumption-free quantity (HB) and whose numerator is exactly the claim being valued; it isolates the residual equity claim without imputing a valuation to the senior instruments, and it keeps the per-share backing legible. Accretive common issuance—sale of equity at a premium to $m^{\mathrm{eq}} = 1$ with proceeds deployed into coins—raises H/N, and it is the long-run trajectory of m^{eq}, rather than the accretion or dilution attributable to any individual issuance, that governs the realized return.

Enterprise value, m^{EV}, aggregates the senior instruments into the numerator and thereby values the entire capital structure relative to its assets. This is a coherent measure of the richness of the whole structure, but it does not isolate the residual claim: it superimposes the lenders’ and preferred holders’ positions on the common holders’, and a single aggregate ratio cannot separate the return accruing to the residual claimant—the object of this analysis—from the cost of the senior financing. It answers a different question than the one posed.

Net asset value, m^{net}, deducts the senior claims from the asset base at face value. This embeds an implicit wind-down assumption, because subtracting $D+L$ at par treats those claims as obligations due and payable in full at the valuation date. For this issuer they are not. The preferred is perpetual: it has no maturity, imposes no date on which coins must be liquidated to redeem principal, and burdens the common solely through its dividend stream—an ongoing carry rather than a principal repayment. The convertibles carry no covenant capable of compelling a sale of Bitcoin; each tranche resolves at its scheduled put either by conversion into equity or by refinancing from ordinary funding. Absent any claim that can force liquidation of the collateral, deducting the full face value of the senior stack prices the equity on wind-down terms even when continued operation is the modal outcome.

This is not to dismiss m^{net}, which is a legitimate downside reference—an estimate of the floor the common would approach were capital-market access to be withdrawn. The objection is to its use as the central estimate. If an investor assigns probability π to a forced wind-down, the liquidation outcome warrants weight π and the going concern weight $1 - \pi$; adopting m^{net} as the headline valuation implicitly sets $\pi = 1$.

The decisive argument for excluding the senior stack from the denominator, and charging it through the cash flows instead, is that the alternative double-counts. Consider the convertibles under an assumed-diluted share count, which presupposes their eventual conversion to equity: that count has already recognized them as prospective dilution of the common. Deducting the convertible principal from the asset base in addition would charge the same claim twice— once as future shares and once as a cash repayment. A convertible resolves as exactly one of the two, conversion (already in the share count) or repayment/refinancing (not in the share count), so a framework that dilutes for conversion must not also subtract the principal. The identical point applies to the perpetual preferred, whose economic burden is the dividend stream the simulation already charges period by period; subtracting its notional in addition would charge for it twice.

The senior claims are therefore not disregarded but charged in the form in which they actually fall due. Preferred dividends, convertible coupons, and any contingent redemptions are deducted as ongoing costs that reduce the asset backing per share, period by period; and the fair premium itself carries a net-backing term equal to the excess of Bitcoin value over the senior claims ranking ahead of the common, normalized by gross coin value, which marks the premium down as that coverage thins—so that a balance sheet on which the coins only narrowly cover the senior stack is valued more conservatively than one with ample headroom. The headline premium thus remains a direct comparison of the market valuation of the common equity to the Bitcoin it ultimately stands behind, with the senior stack borne where it economically falls: in the cash flows, over the holding period.

3 Exogenous state dynamics

Two exogenous drivers are simulated on a quarterly grid: the Bitcoin price and the short policy rate (SOFR). Each admits several processes so that the user can express a range of worldviews rather than a single house view.

3.1 Bitcoin price

Each price model is anchored to its own estimate of *today's fair value* F_0, and the gap between spot and F_0 drives a catch-up. This removes a hidden inconsistency: the power law treats today's price as a discount to a trend line, so it must not be compared against models that silently assume spot *is* fair value. We give each model a fair value computed the way that model would compute one. The process selected here is *your* outlook—it drives the Bitcoin price and hence your return; a second, independently selectable process represents *the market's* outlook and drives the premium the equity trades at (Section 5.2), and the two need not agree.

Power law (default). Following the long-observed log-log linearity of Bitcoin's price against time, the trend is $F_t = A\, g_t^n$ with g_t the number of days since the genesis block, calibrated to $A = 1.6 \times 10^{-17}$ and $n = 5.77$; today's fair value F_0 is read straight off this Santostasi line. The log-deviation $x_t \equiv \ln(B_t/F_t)$ follows a mean-reverting Ornstein–Uhlenbeck process around the line,

$$dx_t = -\kappa_B\, x_t\, dt + \nu(t)\, dW_t^B, \qquad \nu(t) = \sigma(t)\sqrt{2\kappa_B}, \tag{6}$$

with a relative volatility that decays as the network matures, $\sigma(t) = \sigma_0\,(t/t_0)^{-\gamma}$ ($\sigma_0 = 0.50$, $\gamma = 1$). The residual is initialized at today's deviation $x_0 = \ln(B_0/F_0)$, so a price below the line is a discount expected to be recovered—a structurally bullish, low-ruin prior. $B_t = F_t\, e^{x_t}$.

Fair value from the 200-week moving average. The diminishing-returns and GBM models take their fair value from an empirical anchor: the 200-week moving average, WMA_{200}. But the moving average is a *floor*, not a central value—a trailing average of a rising series sits below the series' current level for a purely mechanical reason. We therefore advance it by a model-specific *lag factor*, $F_0 = \phi \cdot \mathrm{WMA}_{200}$, where ϕ equals the current trend value divided by the trend averaged over the trailing 200-week window. A faster-growing trend leaves its trailing average further behind, so the factor is larger for more bullish models: $\phi \approx 1.75$ under the diminishing-returns growth rate and $\phi \approx 1.10$ under the GBM median. With $\mathrm{WMA}_{200} \approx$ \$63k these give fair values near \$110k and \$69k respectively—both well above spot, and both below the power-law line.

Geometric Brownian motion. The skeptic's model: a random walk with constant log-drift μ and volatility σ (defaults 0.20 and 0.55),

$$d\ln B_t = \big(\mu - \tfrac{1}{2}\sigma^2\big)dt + \sigma\, dW_t^B + \delta_t\, dt, \tag{7}$$

to which we add a purely deterministic catch-up drift $\delta_t = -\kappa_B\, x_0\, e^{-\kappa_B t}$ that carries the median from spot up to the fair-value line $F_0\, e^{(\mu-\frac{1}{2}\sigma^2)t}$ over the reversion window while leaving the random-walk dispersion (and its fat left tail) intact. We use the *median* lag factor. For a high-volatility lognormal the mean grows at μ while the typical path grows at $\mu - \frac{1}{2}\sigma^2$ (with $\sigma = 0.55$ the gap is the $\approx 15\%$/yr Itô drag), so anchoring the skeptic's fair value to the mean would import precisely the right-tail optimism the GBM lens exists to exclude. The median—the level half of paths fall below—is the representative central estimate, and it makes the construction internally consistent, since the fair-value line already *advances* at the median rate $\mu - \frac{1}{2}\sigma^2$.

Diminishing returns. A conservative middle ground in which the trend grows from F_0 at a compound rate decaying toward a floor, $c(t) = c_\infty + (c_0 - c_\infty)e^{-\lambda t}$, giving $G_t = F_0 \exp\big[c_\infty t + \frac{c_0 - c_\infty}{\lambda}(1 - e^{-\lambda t})\big]$ around which the same OU residual fluctuates ($c_0 = 0.30$, $c_\infty = 0.08$, $\lambda = 0.10$), with $x_0 = \ln(B_0/F_0)$ so spot catches up to the line exactly as in the power law.

Stochastic volatility (auto-calibrated overlay, on by default). Each process is overlaid with a regime-switching volatility, switched on with no parameter for the user to choose. The instantaneous volatility mean-reverts in log space to the volatility the model already uses—the age-decaying $\sigma(t)$ of the power law, or the constant of the GBM—so the overlay changes the *shape* of the distribution, not the average volatility. Writing $\sigma_t = \bar{\sigma}(t)\, e^{\,y_t - \frac{1}{2}s^2}$ with $dy_t = -\kappa_v y_t\, dt + \xi_v\, dW_t^v$ and $s^2 = \xi_v^2/(2\kappa_v)$ the stationary variance, the $-\frac{1}{2}s^2$ term makes the overlay mean-preserving ($\mathbb{E}[\sigma_t] = \bar{\sigma}(t)$). The two dynamics are set from stylized facts about Bitcoin rather than fit to market data: volatility regimes mean-revert with roughly a three-month half-life ($\kappa_v = 3$) and swing about $\pm 45\%$ in log terms ($\xi_v \approx 1.1$), so volatility occasionally runs to $2.5\times$ or down to $0.4\times$ of its anchor. Because a leveraged balance sheet responds to volatility asymmetrically—a spike can trigger an irreversible forced sale on the downside but, when the firm is strong, can balloon the premium and fund accretive issuance on the upside (Section 7)—turning this overlay on makes the tails both fatter and, under bullish Bitcoin views, net favorable.

3.2 Policy / short rate (SOFR)

The model carries a single rate state, and it is the *short* rate, not the ten-year. The reason is structural and is developed in Section 4.3: STRC is a *variable-rate* perpetual whose dividend the issuer resets frequently against a short benchmark, so the rate that sets its dividend is SOFR—the overnight financing rate the Federal Reserve controls—following the issuer's own "spread-over-SOFR" framing. This is also the rate channel a hawkish "higher-for-longer" stance actually moves. We therefore *forecast SOFR* with one of four processes, jump-diffusion by default to reflect a debt-saturated world in which policy moves in discrete steps on meetings and fiscal news with an upward skew:

$$\text{Vasicek:} \quad dr_t = \kappa(\theta - r_t)dt + \sigma_r\, dW_t^r, \tag{8}$$

$$\text{CIR:} \quad dr_t = \kappa(\theta - r_t)dt + \sigma_r\sqrt{r_t}\, dW_t^r, \tag{9}$$

$$\text{Random walk:} \quad dr_t = \sigma_r\, dW_t^r, \tag{10}$$

$$\text{Jump-diffusion (default):} \quad dr_t = \kappa(\theta - r_t)dt + \sigma_r\, dW_t^r + J_t\, dQ_t, \tag{11}$$

with current SOFR $r_0 = 3.62\%$, long-run neutral $\theta = 3.5\%$, reversion $\kappa = 0.40$ (policy rates adjust faster than long-term yields would), volatility $\sigma_r = 0.9\%$, and a reflecting floor. In the jump variant Q_t is Poisson with intensity 0.15/yr and jumps $J \sim \mathcal{N}(\mu_J, \sigma_J^2)$ with $\mu_J = +50$ bps (fiscal and debt shocks skew rates upward) and $\sigma_J = 150$ bps. A hawkish "higher-for-longer" view is expressed by raising θ (and r_0), a normalization by lowering θ; the shock dW^r may be correlated with dW^B to encode an inflationary regime. There is deliberately no separate long-rate forecast— Section 4.3 explains why a second rate state would mis-model a floating-rate preferred rather than improve it.

Joint fiscal-stress shocks (off by default). Optionally, each Poisson rate jump is treated as a single fiscal-regime event that re-rates Bitcoin in the same instant: when the jump fires, $\ln B$ receives a permanent step $\sim \mathcal{N}(\mu_B^J, (\sigma_B^J)^2)$. With $\mu_B^J > 0$ this encodes the debasement thesis—the event that spikes yields also bids up Bitcoin—while $\mu_B^J < 0$ encodes a risk-off liquidation. It is left off by default because acute fiscal shocks are typically risk-off at the event horizon even if debasement dominates the long run.

A constrained policy-rate reaction function (on by default). The default policy overlay constrains the Fed's *reaction function*: a fiscally-dominant Fed caps how high it lets the policy rate go, and how long it holds it there, once debt service binds. We model this as a gentle, asymmetric pull. The rate evolves freely below a critical value r^*, but above it faces a downward force proportional to the excess, $dr_t -= \lambda_{\text{fd}} \max(r_t - r^*, 0)\, dt$. The force is one-sided and *dormant* below r^*, so at today's sub-r^* rate it does not misstate the present; it engages only when policy would otherwise turn restrictive. It is therefore distinct from the symmetric reversion-to-neutral already in the SOFR process: it is centered *above* neutral and acts only on the upside, encoding an *inability to stay high* rather than a tendency to return to a mean. The default $r^* = 4.5\%$—roughly a percentage point above the 3.5% neutral—is the mildly restrictive stance a debt-constrained Fed is reluctant to exceed, and with $\lambda_{\text{fd}} = 1.0$/yr it leans on the upper tail of the rate distribution: with neutral SOFR near 3.5% it engages only when policy turns restrictive, so its effect on the base case is modest while it does real work in a hawkish or fiscal-dominance scenario. We deliberately do *not* set r^* at the prior cycle's $\approx 5.5\%$ peak, which would imply the Fed could re-scale that peak unencumbered before any constraint binds—the opposite of the fiscal-dominance premise. A hard cap (immediate, dated, or sticky-on-breach) remains available as the limiting case of infinite strength.

4 The capital-structure engine

The novel content of the framework is the propagation of Strategy's actual liability stack, so that b_T and m_T are outcomes of financing decisions rather than free parameters.

4.1 Leverage

Define amplification as senior claims over Bitcoin value,

$$a_t \;=\; \frac{S_t + D_t}{H_t B_t}, \tag{12}$$

where S_t is preferred notional (liquidation preference) and D_t is convertible principal. New issuance targets a chosen amplification a_{tgt}, and a ceiling $a_{\max}$ throttles it; when a drawdown pushes a_t above the ceiling the firm's financing options narrow sharply (below). The base-case assumptions are a target of $a_{\text{tgt}} = 0.45$ and a high boundary of $a_{\max} = 0.60$, against an entry level of ≈ 0.42 implied by the calibration snapshot. There is no hard *low* boundary: in a sustained Bitcoin advance the denominator grows while the senior stack does not, so amplification drifts down toward zero on its own, and the firm re-levers (issuing preferred toward the target) only as the ratio falls back below a_{tgt}. The practical operating band is therefore roughly 0 (deep bull) to 0.60 (stress ceiling), centered on the 0.45 target.

4.2 Endogenous preferred dividend

The preferred stack is split into a fixed-coupon tranche and a floating tranche (STRC). The *required* yield the market demands of STRC is SOFR plus a spread that is the security's inherent baseline plus three *fading stress adders*, each keyed off already-simulated state:

$$q_t^{\text{STRC}} = r_t + \underbrace{s_0}_{\text{baseline}} + \underbrace{k_R\, e^{-w_t/\tau_R}}_{\text{reserve runway}} + \underbrace{k_B\, \max(a_t - \underline{a},\, 0)}_{\text{senior coverage}} + \underbrace{k_V\, \max(\sigma_t^B - \bar{\sigma},\, 0)}_{\text{volatility}} - \rho_{\text{cr}}\, \theta_t - \beta_{\text{coll}} + d_t, \tag{13}$$

floored at $\underline{s} = 3.0\%$. The baseline $s_0 = 5.5\%$ is the spread a junior, Bitcoin-backed, retail-held perpetual commands even when healthy. The *reserve-runway* adder is the central new term: with $w_t = R_t/(\bar{q}_t S_t)$ the years of dividend coverage the USD reserve provides, the adder is large when the runway is short and decays as coverage lengthens ($k_R = 4.0\%$, $\tau_R = 0.6\,\text{yr}$). The *coverage* adder uses the claims senior-through-STRC—convertibles plus the preferred stack—over Bitcoin value, a_t; as converts are repaid or convert to equity, a_t falls and STRC rises in the waterfall, so this adder is also a structural tailwind ($k_B = 16\%$, $\underline{a} = 0.30$). The *volatility* adder ($k_V = 9\%$, $\bar{\sigma} = 0.55$) prices the risk premium in a turbulent tape. The rating-migration term $\theta_t \in [-1, 1]$ seasons upward while the firm pays in full and keeps its reserve and takes a sharp, sticky downgrade on any pause (up to $\rho_{\text{cr}} = 2.5\%$ tightening; $\alpha = 0.35/\text{yr}$, $h = 0.30$); an optional Basel collateral regime adds β_{coll}. The distress term d_t fires *only* in genuine last-resort distress, which the model reaches when three things coincide: (i) the USD reserve is drawn down to its operational floor (a minimum the firm holds back rather than spend, so reserve cash can no longer cover the dividend), (ii) the equity mNAV (m_t) is so depressed that meeting the dividend through the common ATM would be intolerably dilutive (at or below the $0.70\times$ ATM floor), and (iii) the STRC ATM is itself unavailable (below par). In that state the firm does not immediately pause: it sells just enough Bitcoin to cover the dividend, leaving the reserve at its floor, and pauses only if even an orderly per-quarter liquidation (capped at 5% of the stack) cannot meet the bill—a catastrophic-collapse tail. The two resolutions are not equally severe, so the term is tiered: a forced Bitcoin *sale*, which keeps the dividend current, adds 2.0%, while an outright *pause*, which defers the dividend into compounding arrears, adds the full 4.0%. Neither is triggered by the small, discretionary Bitcoin sales Strategy has used to demonstrate process.

The offered coupon ratchets, capped by affordability. Management does not pay the required yield instantly; it resets the dividend toward it in small monthly steps to defend par, but the achievable rate is capped by what the reserve can *afford*:

$$c_t = \operatorname{clip}\Big(c_{t-\Delta} + \operatorname{clip}\big(\min(q_t^{\text{STRC}},\, c_t^{\max}) - c_{t-\Delta},\, -\Delta c,\, +\Delta c\big),\, \underline{c},\, \bar{c}_{\text{will}}\Big), \qquad c_t^{\max} = \operatorname{clip}\Big(\frac{R_t}{\phi\, S^{\text{pref}}},\, \underline{c},\, \bar{c}_{\text{will}}\Big), \tag{14}$$

with a step limit $\Delta c = 0.75\%$ per quarter ($\approx 25\,\text{bps/month}$), an affordability horizon $\phi = 0.6\,\text{yr}$, and a *willingness ceiling* $\bar{c}_{\text{will}} = 15\%$. The ceiling is adopted here as a reasonable modeling assumption rather than a figure attributable to any management statement. The mechanism is deliberately self-limiting: raising the coupon enlarges the dividend bill and *shortens* the runway, which widens the required yield—so beyond a point a higher coupon is counter-productive, and the security absorbs stress in *price* rather than in coupon. With a constant fixed-tranche coupon q^{fix} on notional S^{fix}, the blended preferred coupon is $\bar{q}_t = [S^{\text{fix}} q^{\text{fix}} + (S_t - S^{\text{fix}})c_t]/S_t$.

4.3 STRC price, the reserve, and ATM availability

STRC is engineered as a low-volatility, near-par instrument, and management has stated it will issue new shares through the at-the-market (ATM) program *only at or above par*. Modeling the ATM therefore requires a model of the STRC *price*, not just its required yield—because there are extended periods (the snapshot is one) in which the price trades below par and the ATM is simply shut, even though the security remains over-collateralized.

Which rate drives STRC—and why one forecast suffices. It is worth being precise here, because STRC invites a genuine confusion. By *Macaulay* duration—computed on its cash flows *as if the dividend were fixed*—a perpetual has very long, effectively unbounded, duration, which would seem to make its price acutely sensitive to the *long* rate. But Macaulay duration measures the wrong thing for this security, because STRC is a *variable-rate* perpetual: the issuer resets the dividend frequently (it accrues and pays on a roughly twice-monthly cadence) against a short benchmark. For a frequently-resetting floater, a change in the general level of rates flows through into the *coupon* rather than into the price; its *effective* (rate) duration is short—on the order of the time to the next reset, not the maturity. The frequent dividend cadence is precisely what collapses the effective duration toward the front end. The right benchmark for the *dividend* is therefore the short rate, SOFR, and a one-for-one rise in SOFR raises the required dividend one-for-one (holding the spread fixed). Pricing the dividend off the ten-year instead—treating a floater as a fixed-rate long bond—would be the actual modeling error.

The "long-duration" character does not vanish; it *migrates from rate risk to spread risk*. A perpetual has no maturity to pull its price back to par, so what remains highly sensitive is the *credit/structure spread* the market demands over SOFR (its *spread* duration is long) and the *lag* between a move in the required yield and the issuer's next reset. A Bitcoin-bear, risk-off tape widens that spread regardless of where SOFR sits; a SOFR scare or a vol spike opens a temporary gap before the dividend catches up. Both are captured here—not by a second long-rate state, but structurally, by the pricing identity below: the spread enters through the reserve-runway, coverage, and credit terms, and the reset lag is the small-step ratchet of the coupon toward the required yield. So the two "rate forecasts" a reader might expect collapse into one explicit forecast (SOFR, Section 3.2) for the coupon, plus a long *spread*-duration price risk handled by par defense—rather than a short-rate coupon inconsistently discounted at a long rate.

Price is the dividend over the required yield. A perpetual's price is par scaled by the ratio of the dividend it pays to the yield the market demands,

$$P_t^{\text{STRC}} \;=\; \Pi \cdot \min\Big(\bar{p},\ \frac{c_t}{q_t^{\text{STRC}}}\Big), \qquad \Pi = \$100, \quad \bar{p} = 1.003. \tag{15}$$

The snapshot confirms the identity: a paid coupon of 11.5% against a required yield of $\approx$ 15.5% implies $100 \times 11.5/15.5 \approx \$74$, matching the observed print. So we do not impose a price band; the price is the gap between what Strategy pays (c_t, the ratcheting coupon above) and what the market requires (q_t^{STRC}). The cap $\bar{p}$ encodes *par defense from above*: when STRC would trade at a premium, management trims the coupon (within the reset limit) so the fundamental hugs par rather than floating above it.

The reserve is what the market prices. The dominant term in the required yield is the reserve runway, and the recent history bears this out almost mechanically. Strategy established a USD reserve of \$2.25 B in early 2026—about 2.5 years of dividend coverage—and STRC held its stated amount. A subsequent convertible buyback funded from the reserve cut coverage to roughly six months and STRC slipped to $\approx$ \$97. By mid-2026 the reserve had been rebuilt to $\approx$ \$1.4 B, yet STRC traded near \$74: coverage had actually *improved* from the spring, but Bitcoin had fallen sharply and its volatility spiked, widening the coverage and volatility adders faster than the runway adder narrowed. Over the same window the dividend climbed from 9% toward 11.5% as Bitcoin weakened. The model reproduces this non-monotonicity directly: the reserve sets the regime, but Bitcoin's level and volatility move the price within it. (The structural terms alone do not fully reproduce the spring's resilience—STRC held near par on only six months of coverage—which we read as the market extending confidence that was later withdrawn, rather than as a stable equilibrium.)

The ex-dividend sawtooth and a smooth ATM gate. STRC accretes its (roughly twice-monthly) dividend and drops on the ex-date, with documented selling pressure in the following days. A price defended exactly at par is therefore above par only about half of each cycle. Rather than a hard par gate, the ATM-open fraction is the smooth share of the cycle the price spends at or above par,

$$\text{open}_t \;=\; \text{clip}\!\left(\frac{P_t^{\text{STRC}} - \Pi}{A_{\text{saw}}} + \tfrac{1}{2},\ 0,\ 1\right) \cdot \left[q_t^{\text{STRC}} \le \bar{q}^{\,\text{ceil}}\right] \cdot \left[a_t < a_{\max}\right], \tag{16}$$

with sawtooth amplitude $A_{\text{saw}} = \$2.2$ (the dividend drop plus post-ex-div pressure). Issuance and reserve top-ups scale by this fraction. The behavioral upshot is sharp: even a STRC defended successfully at par funds the ATM only about half the time, so Bitcoin accumulation through it is materially slower than a continuously-issuable instrument would suggest.

The bands are outputs, not assumptions. Because the price is now *derived* from the simulated reserve, senior coverage, and volatility, the time STRC spends near par and the fraction of the horizon its ATM is open are emergent results of each Bitcoin path, not inputs. In the base case the power-law path holds STRC at or above \$97 about 73% of the time with the ATM open about 37%; the diminishing-returns path is similar; and a geometric-Brownian (flat-to-declining) path leaves STRC deeply discounted with the ATM open only about 12% of the time. The reserve is rebuilt toward a thirty-month target via *common* issuance gated on capital-market access (which fades as amplification rises), *not* gated on the STRC ATM. A below-par STRC could never fund the reserve rebuild that lifts it back to par. The engine reports both the fraction of the horizon the ATM is open and the Bitcoin accumulated through it (Section 10).

4.4 Convertible notes

Each convertible tranche is tracked to its put date under an *assumed-diluted* share count: every tranche's conversion shares are pre-counted in N from the outset, consistent with the entry mNAV. At an early call date, if the price is sufficiently above the strike the issuer calls and holders convert—the shares are already in the count, so only the debt is retired. At the put date, if the price is above the strike the notes convert (again no new shares); otherwise the principal is paid in cash through the waterfall *and the pre-counted shares are removed*, so the diluted count falls when dilution does not occur. This keeps entry and exit on the same share basis and correctly credits shareholders for the conversion option expiring worthless. A small blended coupon is carried as a cash drag, and the deep-out-of-the-money tranches dominate forced-cash events in the near-term stress window.

4.5 Financing waterfall

Any cash need—an out-of-the-money convertible principal or a dividend bill that free cash flow cannot cover—is met in strict order:

1. free cash flow, then the cash reserve down to an operating floor;
2. at-the-market *common* equity: accretive when $m > 1$; when m is between a defensive floor m_{floor} (default 0.70, a 30% discount) and 1 the firm still issues to protect the senior stack, diluting the common, capped at a fraction of Bitcoin value per quarter;
3. at-the-market *preferred* (STRC), only while it trades at or above par (Section 4.3);
4. only if m is below the defensive floor or the quarterly cap is exhausted, a forced *sale of Bitcoin*.

The cash (USD) reserve is targeted to a band running from a low boundary of twelve months to a high boundary of thirty months of forward preferred dividends—refilled toward the upper bound in calm periods and drawn down in stress—with a hard operating floor of roughly three months below which the waterfall must resort to selling Bitcoin. The base case starts the reserve at $\approx$ \$1.4 billion. The preferreds carry no forced-liquidation trigger: rather than default, the firm funds the dividend through the waterfall above—including, as a last resort when the reserve is exhausted, common issuance is intolerably dilutive, and the STRC ATM is shut, an orderly sale of Bitcoin sized to cover the bill—and only *pauses* the dividend if even that capped sale falls short, with cumulative series then accruing arrears that compound until cleared.

The defensive floor m_{floor}—default 0.70, meaning the firm keeps selling common to honor its senior commitments down to a thirty-percent discount but no further—is the single most consequential policy lever, trading dilution of the common against forced coin sales. It is worth being explicit about what this issuance does and does not do for the common holder. Below $m = 1$ it is *not* accretive: selling shares for less than the Bitcoin standing behind them lowers Bitcoin-per-share. What it buys is continuation of the strategy. By meeting the dividend and principal obligations with common equity rather than with coins, the firm stays current on its senior commitments and so defends its credit standing—which is precisely what allows the amplification machine to keep running into the next upcycle. Crucially, the senior claims carry no maintenance covenants and no forced-liquidation trigger, so below-par issuance is a discretionary cost the firm elects to bear in order to protect its standing, never an obligation that could compel asset sales at the bottom of a drawdown. Put plainly, the common holder accepts some Bitcoin-per-share dilution in the bad times as the regrettable but bounded price of preserving the amplification that delivers outsized Bitcoin-per-share accretion in the good times. The model treats the level of this floor as a policy choice rather than a fixed constant, since reasonable people will weigh that trade differently.

Proactive accretive issuance. The waterfall above is defensive—it raises cash only to meet an obligation. The firm also issues *offensively*: when the premium is present ($m \geq 1.05$, so issuance is accretive) and Bitcoin is in strength (above a trailing reference), it sells common equity and buys

Bitcoin, raising Bitcoin-per-share. Issuing ΔN shares at premium m changes backing per share by a factor $1 + \Delta N(m-1)/(N + \Delta N)$, accretive precisely when $m > 1$. Crucially the pace *scales with the premium*: the quarterly issuance is $\min\big(\eta\,(m-1),\, \bar{a}\big)$ as a fraction of Bitcoin value, with slope $\eta = 0.10$ and a market-depth cap $\bar{a} = 12\%$. At a 1.1× premium this is a trickle; at a 2–3× premium the firm drives a truck through the window. Because the proceeds buy coins, this program *de-levers* (it raises Bitcoin value without adding senior claims), and it is the mechanism through which a volatility-driven premium spike is converted into permanent Bitcoin-per-share.

5 The premium (mNAV) function

5.1 The structural premium

At any point the *structural* premium the common commands comprises three components—a net-backing ratio, an option/convexity premium, and a convex upside flywheel,

$$s_t \;=\; \underbrace{\frac{H_tB_t - L_t + R_t}{H_tB_t}}_{\text{net-backing ratio } \rho_t} \;+\; \underbrace{\Big(\pi_{\text{floor}} + \pi_{\text{cvx}}\tfrac{\sigma_t}{\sigma_{\text{ref}}}\Big)}_{\text{option/convexity premium } \pi_t} \;+\; \underbrace{\pi_{\text{kick}}\left[\max\!\big(\tfrac{\sigma_t}{\bar{\sigma}_t} - 1, 0\big)\right]^2 \widehat{\rho}_t}_{\text{convex upside flywheel}}, \tag{17}$$

where L_t is senior liabilities including arrears, R_t the reserve, and $\widehat{\rho}_t = \text{clip}(\rho_t, 0, 1)$. Each component is explained in turn; the following subsection then specifies *whose* beliefs evaluate it and how the price the investor actually pays relates to it.

Net-backing ratio ρ_t. The first component anchors the premium to the Bitcoin that backs the common *after* the senior stack is satisfied: gross coin value H_tB_t, less senior liabilities L_t (preferred par, accrued dividend arrears, and convertible principal), plus the cash reserve R_t, all expressed per unit of gross coin value. It is the level to which the premium would collapse in the absence of any going-concern optionality, and it encodes the fact that the common is a *residual* claim: leverage levers its claim on the coin, so a balance sheet on which the coins only thinly cover the senior stack carries a low ρ_t and a correspondingly compressed premium. This same term is what keeps the *exit* valuation honest about the obligations a future buyer inherits. A buyer at the horizon purchases a junior claim on a company that must continue servicing the senior stack—most importantly the perpetual STRC dividend—and the model credits the common only with the coins net of those claims. For a perpetual preferred trading near par at a market-clearing coupon, subtracting its par from the asset base is precisely the capitalized value of the dividend stream the buyer inherits, since par equals that dividend divided by the required yield; the deduction is therefore the present value of an indefinite obligation, not a liquidation haircut. The burden enters through two non-overlapping channels: dividends paid *before* exit have already reduced the backing per share, charged period by period in the waterfall, while the dividends owed *after* exit are capitalized into L_T at the moment of sale. The buyer thus pays for the coin behind each share net of the senior claims, never the gross coin as though the seniors did not exist.

Option/convexity premium π_t. The second component is the premium the common commands *above* its net backing for the option-like character of a levered Bitcoin balance sheet. A levered, long position in a volatile, trending asset has a convex payoff: the upside is effectively unbounded while the downside is capped at the residual claim, and that asymmetry is worth a premium that rises with volatility. We represent it as a floor $\pi_{\text{floor}} = 0.15$ plus a volatility-scaled component governed by $\pi_{\text{cvx}} = 0.30$ acting through $\sigma_t/\sigma_{\text{ref}}$, the prevailing Bitcoin volatility relative to a reference level.

The term is suppressed when the preferred dividend is paused: a structure visibly in distress does not command an optionality premium.

Convex upside flywheel. The third component corrects an asymmetry the smooth convexity premium misses. The largest historical premium expansions occur in *volatility spikes on otherwise healthy balance sheets*, when management can issue aggressively into strength and convert that volatility into permanent Bitcoin-per-share. The term activates only when realized volatility exceeds its own trailing anchor $\bar{\sigma}_t$, scales with the *square* of that excess—so it is dormant in calm markets and explosive in genuine spikes—and is multiplied by the clipped strength $\widehat{\rho}_t$ so that it fires only on well-collateralized paths. A volatility spike on a strong path therefore balloons the premium toward the 2–3× levels the security has historically reached, whereas a high-volatility *crash*—where ρ_t is low—leaves the term near zero and lets the premium compress through ρ_t as it should. With kick coefficient $\pi_{\text{kick}} = 1.0$, the component is convex (squared) in the volatility excess and vanishes identically when the stochastic-volatility overlay is off.

5.2 Two frames: whose beliefs set the price, whose set the premium

A single investor's beliefs about Bitcoin do not move the premium the market grants the equity; that premium is set by the market in aggregate. The framework therefore separates two frames. *Your* Bitcoin outlook—the price process of Section 3.1—drives the simulated Bitcoin price, and through it the backing per share, the coin appreciation, and ultimately your return. *The market's* Bitcoin outlook—a second, independently selectable price process—drives the premium the stock trades at.

A preliminary *market pass* simulates the structural premium s_t under the market's outlook and records its cross-path median at each step, producing a single deterministic trajectory m_t—the *market's rational mNAV path.* It begins at the structural floor $m_0 = \rho_0 + \pi_0$ (the leverage-dragged entry premium, near 1.06× for the base balance sheet, since the heavy preferred-plus-convertible stack thins ρ_0) and evolves as the market's outlook lets the backing improve. The investor's own passes then take this path as given: the mNAV the stock trades at follows m_t, with the observed entry quote reverting toward it,

$$m_t \;=\; m_t \;+\; \underbrace{g_0\, e^{-\kappa_\pi t}}_{\text{entry dislocation}}, \qquad g_0 \equiv m_0 - m_0, \tag{18}$$

the dislocation decaying at $\kappa_\pi = 0.25/\text{yr}$ so the simulation can begin from today's actual quote—a discount or premium that need not equal the market's structural level on day one—without imposing it in perpetuity.

Two consequences of this construction are worth stating, because they fix what the investor's beliefs can and cannot do to the answer. *First, the premium path is deterministic.* Because m_t is the *median* structural premium recorded under the market's outlook, it is one fixed trajectory, applied identically to every Bitcoin scenario in the investor's pass; it does not re-respond to the realized Bitcoin price within a given path. All of the scenario-to-scenario dispersion in the exit value therefore lives in the backing per share b_T, while the premium at which the share is sold is fixed by the market. In the intrinsic (fair-value) pass the dislocation is zero, so the mNAV *is* the market path exactly; in the observed pass the reverting gap is present, but that pass feeds only the return distribution, not the rational mNAV. *Second, the market's outlook shapes backing growth, not only the exit multiple.* The accretive equity program issues against the prevailing mNAV—and that mNAV is the market's path m_t, not the investor's—so the flywheel runs at the market's premium.

The market's outlook therefore governs both the multiple at which the investor exits *and*, through the pace of accretive issuance, how much Bitcoin-per-share is accumulated over the hold. Stated cleanly: the investor's outlook sets the Bitcoin price—and so the coin behind each share and its appreciation—while the market's outlook sets the premium and, through the flywheel, modulates the growth in backing. The investor's bullishness cannot inflate the premium; it can only earn more coins per share, which are then marked at the market's multiple.

6 Finite-supply accumulation brakes

The capital-structure engine, left unconstrained, would let the firm issue and accumulate Bitcoin without bound—incompatible with a fixed 21 million-coin supply. Rather than impose a terminal holding as an input, the model lets one *emerge* from three economically grounded negative feedbacks, each keyed to the firm's share of accessible supply,

$$\theta_t \;=\; \frac{H_t}{S_{\text{acc}}}, \qquad S_{\text{acc}} = 17.3\,\text{million BTC}, \tag{19}$$

where S_{acc} is the 21 million cap less coins presumed permanently lost. As θ_t rises the three brakes tighten, and the terminal stack asymptotes as a *derived* output whose level depends on beliefs.

(1) Float-depletion slippage. A large buyer is a price-*taker* on the global Bitcoin trend but a price-*mover* on its own executions: it must walk a thinning order book. The chosen Bitcoin process therefore remains the market price B_t—the numeraire, the mark, and the mNAV denominator—and is never overwritten; finite supply enters only as a slippage markup on Strategy's *own* marginal purchases,

$$c_t \;=\; B_t\Big(1 + \underbrace{k_{\text{perm}}\,\tfrac{\theta_t}{1-\theta_t}}_{\text{permanent scarcity}} + \underbrace{k_{\text{temp}}\,\tfrac{P_t/B_t}{S_{\text{acc}}-H_t}}_{\text{temporary, buy-size}}\Big), \tag{20}$$

so a financing inflow P_t buys P_t/c_t coins. Both terms diverge as $H_t \to S_{\text{acc}}$, so the brake binds on *quantity*: as the float thins each dollar buys fewer coins and the stack approaches the float asymptotically, while B_t continues to follow whatever model the user selected. Reflexive "buying lifts the price" is deliberately excluded—the power law already embeds historical demand.

(2) Concentration-driven premium decay. The optionality component of the premium—π and the volatility flywheel of the structural premium—is the capitalized value of *future* accretion. As the firm absorbs the float that future shrinks, so the premium above backing should decay toward zero. The optionality is therefore scaled by

$$g_t^{\text{conc}} \;=\; \left(\tfrac{1-\theta_t}{1-\theta_0}\right)^{p_{\text{conc}}}, \tag{21}$$

equal to one at today's concentration θ_0 and falling to zero as $\theta_t \to 1$. At saturation the premium collapses to the net backing ratio ρ_t, at which point the accretive common ATM—which issues only above m_{acc}—self-halts. The channel is visible in the premium path itself: under a trend outlook the market's structural premium first rises, then *declines* at long horizons as concentration compresses it.

(3) STRC carry closure. Preferred-funded accumulation only makes sense while the asset outearns the coupon. Proactive STRC issuance toward the amplification target is therefore gated on

$$g^{\mathrm{BTC}}(t) \;>\; c_t^{\mathrm{STRC}} + \mathrm{margin}, \tag{22}$$

where $g^{\mathrm{BTC}}(t)$ is the local forward growth of the selected Bitcoin model—for the power law, $\alpha/(t_0 + t)$, declining with age. As that growth decays toward the prevailing coupon, or a distressed coupon rises toward it, the carry closes and the STRC channel stops adding coins.

An emergent, belief-dependent terminal stack. Together the three brakes replace an arbitrary terminal holding with a derived asymptote. Under the default beliefs (your power law, the market's diminishing returns) the median stack climbs from $\approx 1.5\,\mathrm{M}$ coins at ten years to $\approx 4.0\,\mathrm{M}$ (23% of accessible supply) at twenty-one years, then converges to $\approx 8.8\,\mathrm{M}$ ($\approx 51\%$) only over forty-to-eighty-year horizons; a control with the brakes disabled diverges without bound across the same spans. The asymptote is belief-dependent—a more conservative Bitcoin view accumulates less—so it is properly an output of the analysis rather than a dial. Because that level is reached only at horizons where the framework is least reliable, the long-run figures are best read qualitatively.

7 Fair value and the rational mNAV

Each simulated path produces an exit value for a single share, expressed in Bitcoin: $v_T \equiv b_T\, m_T$, the product of the Bitcoin backing per share at the horizon, $b_T = H_T/N_T$ (coins held divided by diluted shares), and the terminal premium m_T. For the fair-value calculation the entry dislocation is set to zero, so this terminal premium is the market's path endpoint m_T of Section 5.2—the level the market grants, not one inflated by the investor's own outlook—while b_T carries the investor's Bitcoin outlook. A share bought today for V_0 dollars costs V_0/B_0 coins at the current spot price B_0, and the buyer is compensated for the wrapper's incremental risk only if the median terminal coin value clears that entry cost grown at the hurdle. Equating the two and solving for the fair dollar price gives

$$V_0 \;=\; B_0\,\frac{\mathrm{med}(v_T)}{(1+r_p)^{\tau}}, \tag{23}$$

where V_0 is the fair MSTR price today, B_0 the current Bitcoin spot price, $\mathrm{med}(\cdot)$ the cross-path median, r_p the annual counterparty/wrapper hurdle, and τ the holding horizon in years. The factor $\mathrm{med}(v_T)/(1+r_p)^{\tau}$ is the present value, in coin, of the median exit share; multiplying by spot B_0 re-expresses that coin value as a dollar price. This choice of the median is not arbitrary: at a price set by (23), a path beats self-custody exactly when v_T exceeds its own median, which by construction happens with probability one-half. *The fair value is the price at which the probability of beating self-custodied Bitcoin equals* 50%—a self-consistent definition we verify numerically (re-simulating at V_0 returns a beat probability of 0.500). Translating to the premium of this paper,

$$m_0^{\star} \;=\; \frac{V_0\,N_0}{H_0 B_0} \;=\; \frac{\mathrm{med}(v_T)}{b_0\,(1+r_p)^{\tau}}, \tag{24}$$

the *rational mNAV*: the premium at which a marginal buyer with these beliefs is indifferent between the equity and the coin. A firm that merely held Bitcoin, never issued accretively, and exited at parity would have $m_0^{\star} = (1+r_p)^{-\tau} < 1$—a structural discount that compensates for wrapper risk. Every point of premium above one is, in this framework, a quantified bet that the accretion flywheel and the terminal premium will outrun that discount.

Input	Value (snapshot)
Bitcoin held	847,363 BTC
Bitcoin spot price	$\approx$ \$60,000
Bitcoin 200-week moving average	$\approx$ \$62,850
Bitcoin value of treasury	$\approx$ \$50.8 B
MSTR share price	$\approx$ \$92.68
Entry mNAV (this paper's definition)	$\approx 0.73\times$
Diluted shares	398,721,017
Preferred liquidation preference (5 series)	$\approx$ \$15.5 B
STRC share of preferred (floating)	$\approx$ 68% ($\approx$ \$10.5 B)
STRC price (snapshot)	\$83.67 ($\approx$ 16% below par)
STRC coupon / implied required yield	12% / $\approx$ 14.3%
Convertible principal (6 tranches)	$\approx$ \$6.71 B
USD reserve	$\approx$ \$2.55 B
SOFR / policy rate	$\approx$ 3.62%
Counterparty/wrapper hurdle r_p (base)	3.5%/yr

Table 1: Base-case calibration snapshot.

The market's rational mNAV versus your personal one. Two distinct premia emerge, and they answer different questions. The *market's* rational mNAV is the path m_t itself—what the equity trades at, set by the market's outlook, running from the structural floor $m_0 \approx 1.06\times$ to its horizon endpoint m_T. The investor's *personal* rational mNAV is $m_0^\star$ above: the entry premium at which an investor holding their own Bitcoin outlook and hurdle is indifferent, given that the stock will exit at the market's m_T. The two coincide only when the investor's beliefs match the market's; a more bullish outlook than the market's lifts $m_0^\star$ above m_0—one would rationally pay more than today's structural premium because one expects more coins behind each share by exit—even though the position is still marked at the market's multiple at sale. Because the fair-value pass holds the premium at the market path (dislocation zero), $m_0^\star$ is independent of today's quoted price: re-feeding the fair price V_0 back as the input leaves it unchanged, so the rational mNAV is a function of beliefs and the balance sheet alone, not of the current market mispricing.

8 Calibration

Table 1 lists the snapshot used to anchor the base case. These are point-in-time inputs and are expected to move; the framework is designed to be re-calibrated rather than to assert fixed values.

8.1 Per-instrument capital structure

The aggregate convertible and preferred figures in Table 1 resolve into the individual instruments the engine carries. Table 2 lists the six convertible tranches—each resolved contingently at its holder put date under the assumed-diluted share count—and Table 3 the five preferred series. Of these, only STRC is issued further and repriced over the simulation; the four fixed preferred series and all six convertibles are held at their snapshot terms.

Tranche	Principal ($M)	Coupon	Conversion px	Investor put (yrs)
2028	1,010	0.625%	$183.19	1.2
2029	1,500	0%	$672.40	2.0
2030 A	800	0.625%	$149.77	2.2
2030 B	2,000	0%	$433.43	1.7
2031	604	0.875%	$232.72	2.2
2032	800	2.25%	$204.33	3.0
Total	6,714			

Table 2: Convertible note tranches at the snapshot (assumed-diluted basis). "Investor put" is the approximate time to the next holder put date, the model's resolution trigger; conversion prices are per MSTR share.

Series	Liq. pref. ($M)	Coupon	Rate type
STRF	1,284	10%	fixed
STRK	1,402	8%	fixed, convertible
STRD	1,402	10%	fixed
STRE	883	10%	fixed
STRC	10,489	12%	variable (modeled dynamically)
Total	15,460		

Table 3: Preferred series at the snapshot. The four fixed series carry a par-weighted blended coupon of $\approx 9.44\%$; STRC, the largest tranche, carries the floating coupon the engine resets each period off SOFR, coverage, and the migrating credit rating.

9 Default assumptions

The snapshot above fixes the *market* inputs. The model also ships with a set of *behavioral* and *policy* defaults that govern how the capital structure responds to those inputs—how aggressively the firm issues, how it defends STRC near par, how it manages its reserve, and which thesis channels are active. Table 4 states them in one place. They are modeling choices rather than observables, and because several of them move the answer materially, they are listed explicitly so a reader can see exactly what the base case assumes and change any of it in the tool.

A handful of these carry most of the sensitivity, and it is worth being plain about which. The *Bitcoin model*, the *market's outlook*, and the *hurdle* dominate the headline: the investor's outlook and the market's outlook jointly set the spread reported below—the former through the backing accumulated, the latter through the premium the price follows—and the premium falls monotonically as r_p rises. The *stochastic-volatility flywheel* mainly shapes the *market's* premium path; in the two-frame split its residual effect on the personal valuation is modest and, at the default, slightly negative (the median's volatility drag outweighs the lost premium kick). The *reserve band* and the *defensive mNAV floor* together govern behavior in stress: a higher floor trades more common dilution for fewer forced coin sales, and a thicker reserve postpones both. The *STRC band targets* and the *par gate* set how often the preferred ATM is open, and therefore how much Bitcoin the flywheel can accumulate. The three *thesis toggles* that ship off—a joint fiscal-shock channel, a hard rate cap, and Basel collateral eligibility—are deliberately conservative; switching them on mainly reshapes the tails. Every figure in the next section is conditional on the values above.

Assumption (symbol)	Default
Decision frame	
Holding horizon (τ)	10 years
Counterparty / wrapper hurdle (r_p)	3.5%/yr
Bitcoin model (your outlook)	power law
Market's Bitcoin outlook	diminishing
Monte Carlo paths $\times$ step (n_p)	20,000$\times$ quarterly
Reserve and financing waterfall	
USD reserve, start (R_0)	$\approx$ \$2.55 B
Reserve refill band (fwd preferred divs)	12 mo / 30 mo
Reserve hard operating floor	3 months
Defensive common-ATM floor (m_{floor})	0.70
Below-par common issuance cap	3%/qtr of BTC value
Last-resort BTC sale (fund dividend) / per-qtr cap	on / 5% of stack
Structural mNAV floor ($m_{\min}$)	0.30
Accretive common-ATM trigger (m_{acc})	$m \geq 1.05$
STRC pricing (required yield → coupon → price)	
Baseline spread / floor (s_0, $\underline{s}$)	5.5% / 3.0%
Reserve-runway adder (k_R, τ_R)	4.0%, 0.6 yr
Coverage adder (k_B, $\underline{a}$)	16%, 0.30
Volatility adder (k_V, $\bar{\sigma}$)	9%, 0.55
Distress add — pause / forced BTC sale (d)	4.0% / 2.0%
Coupon step / willingness ceiling (Δc, $\bar{c}_{\mathrm{will}}$)	0.75%/qtr / 15%
Affordability horizon (ϕ)	0.6 yr
Price cap / sawtooth amplitude ($\bar{p}$, A_{saw})	1.003 / \$2.2
Rate ceiling ($\bar{q}^{\,\mathrm{ceil}}$)	16%
Amplification (leverage policy)	
Target / ceiling / floor ($a_{\mathrm{tgt}}, a_{\max}, a_{\min}$)	0.45 / 0.60 / 0.30
STRC issuance pace cap	5%/qtr of BTC value
Finite-supply accumulation brakes	
Accessible supply (S_{acc})	17.3 M BTC
Float-depletion impact ($k_{\mathrm{perm}}, k_{\mathrm{temp}}$)	0.25 / 2.0
Premium concentration decay (p_{conc})	1.0
STRC carry margin	0%
Short policy rate (SOFR)	
Neutral / long-run (θ)	3.50%
Mean-reversion speed / vol (κ, σ_r)	0.40/yr / 0.90%
Jump mean / intensity (μ_J, λ_J)	+50 bps / ∼0.15 yr^{-1}
Policy-rate cap threshold (r^*)	4.5%
Bitcoin dynamics	
Power-law exponent (α)	5.77
Initial residual vol (σ_0, decaying)	0.50
Stochastic-vol overlay	on
Diminishing / GBM lag factors ($\phi_{\mathrm{dim}}, \phi_{\mathrm{gbm}}$)	1.75 / 1.10
Premium (mNAV) function	
Durable / convex premium ($\pi_{\mathrm{floor}}, \pi_{\mathrm{cvx}}$)	0.15 / 0.30
Flywheel kick / dislocation decay ($\pi_{\mathrm{kick}}, \kappa_\pi$)	1.0 / 0.25 yr^{-1}
Thesis toggles	
On by default	jump rates, vol flywheel, credit migration, policy-rate cap
Off by default	joint fiscal shock, hard rate cap, Basel eligibility

Table 4: Behavioral and policy defaults—modeling choices, not market data.

10 Results

All figures below use the default profile: jump-diffusion rates, the stochastic-volatility flywheel, credit-rating migration, and the policy-rate cap all active, each Bitcoin model anchored to its own fair value, and the three finite-supply brakes engaged. The optional thesis levers (joint fiscal shocks, hard yield-curve control, Basel collateral eligibility) remain off.

Two frames, two readings. The model reports a *market* rational mNAV and a *personal* one. At the default—the market's diminishing-returns outlook, a ten-year hold, and the 3.5% hurdle—the market's structural premium runs from 1.06× at entry (the leverage-dragged floor, with the heavy preferred-plus-convertible stack thinning net backing) to ≈ 1.01× at exit: the level the price reverts to and follows over the hold. Your own Bitcoin outlook then drives the accretion that earns a *personal* rational mNAV around that path—about 1.38× under the power law, 1.32× under diminishing returns, and 0.52× under GBM.

The Bitcoin process dominates. That spread is the entire mNAV debate compressed into one assumption: whether Bitcoin reliably appreciates and recovers from a discount. Because each model carries its own fair-value catch-up, even the conservative diminishing view sits well above net asset value—it, too, reads today's price as a discount to a lag-corrected trend. Under GBM the same leverage and dividend obligations cannot be out-earned by weak growth, and there is no compounding premium to rescue the equity, so backing per share erodes and the rational price is a steep discount: holding the coin wins decisively.

The wrapper hurdle. Holding the power law and a ten-year hold, the personal rational mNAV falls monotonically with the hurdle: 1.95× at 0%, 1.38× at the 3.5% default, and 1.09× at 6%. A sufficiently skeptical view of the *structure* alone compresses the premium toward parity even for a Bitcoin bull; conversely, the wrapper-premium investor of Section 2 who applies a negative hurdle lifts it further—to ≈ 2.15× at −1%—the rational counterpart of paying above Bitcoin backing for convenience, trust, and the avoidance of an ETF fee.

What the default profile contributes. The two-frame construction changes the decomposition materially. Because the optionality flywheel now expresses itself in the *market's* premium path—which the personal pass takes as given—its residual effect on the personal fair value is through dispersion rather than premium: disabling stochastic volatility *raises* the personal mNAV slightly (from 1.38× to 1.48×), as the median benefits from the removed volatility drag more than it loses from a quieter premium. Credit-rating migration is mildly accretive in the other direction (disabling it lowers the figure to 1.33×), since seasoning compresses the STRC coupon and cheapens the accretive channel. The policy-rate cap is essentially neutral at the default, engaging only in the restrictive-policy tail.

STRC availability and accumulation. Because the STRC price is *derived* from the simulated reserve, senior coverage, and volatility, both *when* and *how much* are outputs of each path. At the default, STRC sits within a dollar of par about 69% of the time with the ATM open about 35% of the horizon, and the median path accumulates Bitcoin via the STRC channel equal to roughly 100% of the starting stack over ten years. A flat-to-declining path leaves STRC discounted and funds far less. The binding constraint is the reserve and the coverage it backs, *not* the level of rates—reproducing the snapshot's lived experience, in which a Bitcoin drawdown and a drawn-down reserve, not a rate scare, froze the ATM while STRC remained over-collateralized.

The belief landscape. With the market's outlook now a separate axis, the belief space is a $3 \times 3 \times 3 \times 3$ grid over the market's model, your model, the hurdle (0/3/6%), and the horizon (7/14/21 years). Figures 1–3 and Tables 5–7 present it as three cubes, one per market outlook, each sliced into three panels by your own model. Three patterns stand out. First, the two Bitcoin axes dominate: the bullish corners sit well above net asset value and the GBM corners well below, with financing details a secondary influence. Second, horizon now cuts both ways—a longer hold lifts the bullish cells through accretion but, past roughly a decade, the finite-supply brakes and the concentration-compressed premium pull the most aggressive long-horizon cells back down rather than letting them run. Third, the market's outlook matters most through the premium it grants: when the market reads Bitcoin as GBM, its structural premium collapses toward the backing ratio, dragging even a personal bull's valuation down relative to the same beliefs under a trend-following market.

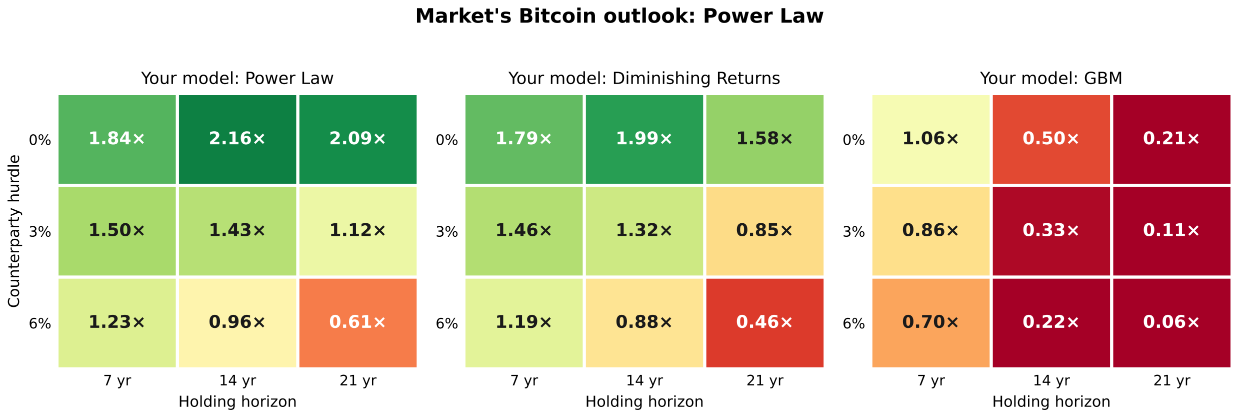

Figure 1: Personal rational mNAV when the market's Bitcoin outlook is Power Law. Panels are *your* Bitcoin model (power law, diminishing returns, GBM); within each panel, columns are the holding horizon (7/14/21 years) and rows the counterparty hurdle (0/3/6%). Green denotes a premium to net asset value, red a discount; the dividing line is 1.0×.

	Counterparty hurdle		
Your Bitcoin model	0%	3%	6%
7-year hold			
Power Law	1.84×	1.50×	1.23×
Diminishing returns	1.79×	1.46×	1.19×
GBM	1.06×	0.86×	0.70×
14-year hold			
Power Law	2.16×	1.43×	0.96×
Diminishing returns	1.99×	1.32×	0.88×
GBM	0.50×	0.33×	0.22×
21-year hold			
Power Law	2.09×	1.12×	0.61×
Diminishing returns	1.58×	0.85×	0.46×
GBM	0.21×	0.11×	0.06×

Table 5: Personal rational mNAV across the 27 belief sets when the *market's* Bitcoin outlook is power law (default profile).

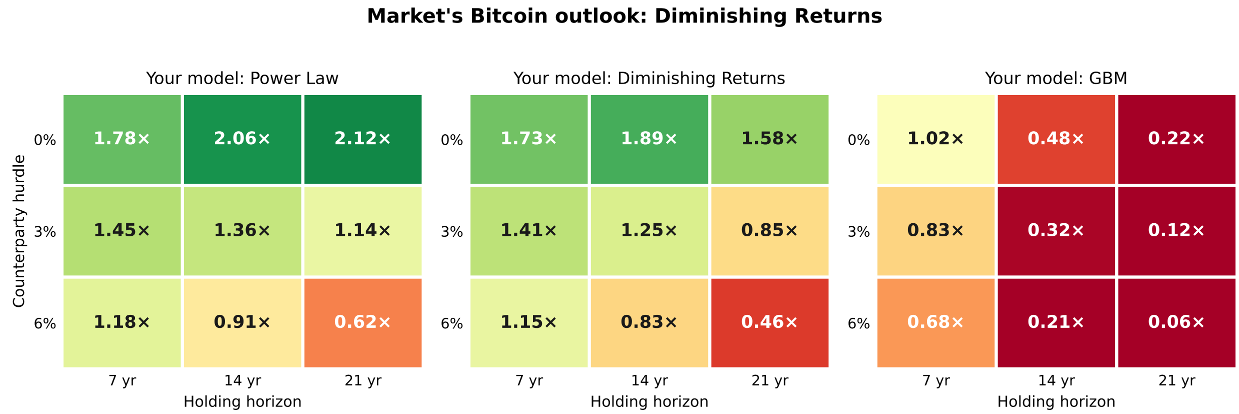

Figure 2: Personal rational mNAV when the market's Bitcoin outlook is Diminishing returns. Panels are *your* Bitcoin model (power law, diminishing returns, GBM); within each panel, columns are the holding horizon (7/14/21 years) and rows the counterparty hurdle (0/3/6%). Green denotes a premium to net asset value, red a discount; the dividing line is 1.0×.

	Counterparty hurdle		
Your Bitcoin model	0%	3%	6%
7-year hold			
Power Law	1.78×	1.45×	1.18×
Diminishing returns	1.73×	1.41×	1.15×
GBM	1.02×	0.83×	0.68×
14-year hold			
Power Law	2.06×	1.36×	0.91×
Diminishing returns	1.89×	1.25×	0.83×
GBM	0.48×	0.32×	0.21×
21-year hold			
Power Law	2.12×	1.14×	0.62×
Diminishing returns	1.58×	0.85×	0.46×
GBM	0.22×	0.12×	0.06×

Table 6: Personal rational mNAV across the 27 belief sets when the *market's* Bitcoin outlook is diminishing returns (default profile).

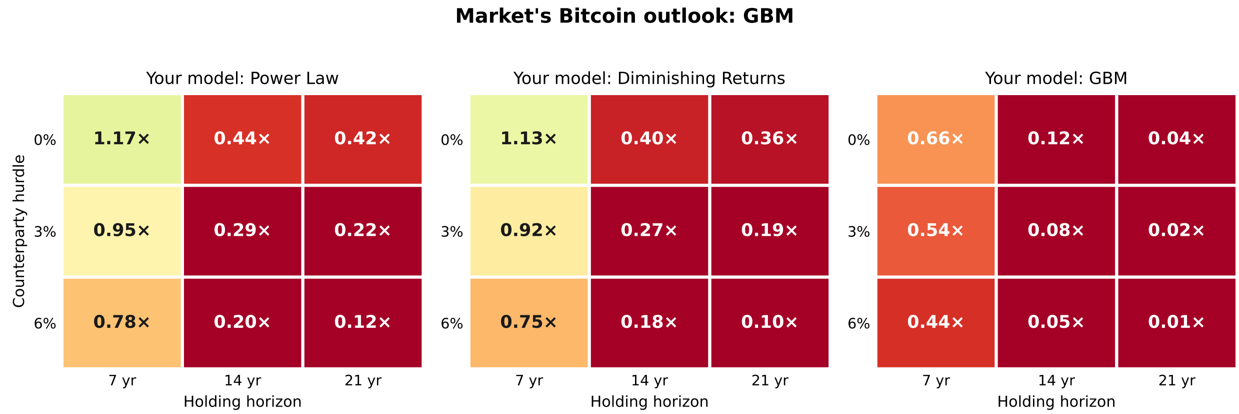

Figure 3: Personal rational mNAV when the market's Bitcoin outlook is GBM. Panels are *your* Bitcoin model (power law, diminishing returns, GBM); within each panel, columns are the holding horizon (7/14/21 years) and rows the counterparty hurdle (0/3/6%). Green denotes a premium to net asset value, red a discount; the dividing line is 1.0×.

	Counterparty hurdle		
Your Bitcoin model	0%	3%	6%
7-year hold			
Power Law	1.17×	0.95×	0.78×
Diminishing returns	1.13×	0.92×	0.75×
GBM	0.66×	0.54×	0.44×
14-year hold			
Power Law	0.44×	0.29×	0.20×
Diminishing returns	0.40×	0.27×	0.18×
GBM	0.12×	0.08×	0.05×
21-year hold			
Power Law	0.42×	0.22×	0.12×
Diminishing returns	0.36×	0.19×	0.10×
GBM	0.04×	0.02×	0.01×

Table 7: Personal rational mNAV across the 27 belief sets when the *market's* Bitcoin outlook is GBM (default profile).

For reference, the snapshot's observed $\approx 0.73\times$ entry mNAV sits below every trend-model cell and inside the GBM range—it is "fair" only under a thoroughly skeptical view of Bitcoin. Under either trend model, across the grid, the equity reads as cheap relative to self-custody.

11 Limitations

It is a capital-structure model, not a macroeconomic one. The engine has exactly two exogenous drivers—the short policy rate (SOFR) and the Bitcoin price—and everything else is the liability stack propagating off them. Broad macroeconomic conditions such as global liquidity, the growth of money supply, national debt and deficits, and reserve-currency status are *not* state variables and are never modeled directly. They enter the analysis only to the extent that the user believes they are already expressed in those two paths: a view that abundant liquidity and monetary debasement lie ahead is encoded by choosing a more bullish Bitcoin process and a more constrained (capped) policy-rate process, not by any separate macro module. The model inherits whatever macro thesis the inputs imply and no more.

The model does not assume fiscal dominance—the user selects it. This is worth stating plainly, because the default profile leans toward a fiscal-aware world and could be mistaken for a built-in assumption. It is not. Whether the simulated future resembles *fiscal* dominance (rates pinned below inflation while hard assets appreciate) or *monetary* dominance (a central bank free to hold real rates positive and anchor inflation) is determined entirely by the settings. Jump-diffusion rates with the policy-rate cap engaged and a power-law Bitcoin trend describe a fiscal-dominance scenario; Vasicek rates reverting to a stable mean with the cap off, paired with a GBM or diminishing-returns Bitcoin view, describe an orthodox monetary-dominance world in which the wrapper usually loses to self-custody. The joint-fiscal-shock and yield-curve-control toggles sharpen the fiscal case further, and all of them ship *off* except the policy-rate cap, which is dormant at today's rates. The framework therefore takes no stance on the debate; it prices whichever side the user assumes.

The STRC price is derived; its calibration is assumed. Unlike an earlier version of this model, the price-stability bands of Section 4.3 are no longer an input: STRC's price, the fraction of time it trades near par, and its ATM-open duty cycle are *emergent outputs* of the pricing identity, driven by the simulated reserve runway, senior coverage, and volatility. What remains a *modeling assumption* is the calibration of the required-yield spread—the inherent baseline and the three adder coefficients—together with the coupon affordability logic, its 15% willingness ceiling (adopted as a reasonable figure, not one attributable to management), the small-step reset cadence, and the ex-dividend sawtooth amplitude. These were set against the instrument's short but informative history: at par on ≈ 2.5 years of reserve coverage in early 2026, near \$97 on roughly six months, and near \$74 on roughly ten months amid a Bitcoin drawdown—so the model reproduces both the level and the *non-monotonicity* (more coverage, lower price, when Bitcoin fell). One gap is acknowledged: the spring's resilience—STRC holding near par on only six months of coverage—is read as confidence the market later withdrew rather than a stable equilibrium, so the structural terms price it somewhat below the observed \$97–98. What still lies in the tail is an outright *breakdown* of the stabilization mechanism (demand evaporates, the reserve genuinely exhausts, and STRC re-rates to a distressed yield), which the distress term captures only on actual reserve exhaustion. These coefficients will need revision as more of the instrument's behavior is observed.

Two technical simplifications. Nominal and real rates are not separated, so a pure debasement that lifts Bitcoin and nominal rates together is captured only approximately, through the optional rate–Bitcoin correlation and joint shock rather than through an explicit inflation state. And the exit value depends weakly on the entry price through the dislocation term, so the fair value of (23) is a one-pass solution rather than a fixed point—though the residual error is small because the dislocation decays well before a multi-year horizon.

It is conditional by design. The framework takes the user's beliefs as inputs and returns the internally consistent premium. Its purpose is to make a valuation follow rigorously from a worldview, not to tell the user which worldview is correct—the two largest uncertainties, which Bitcoin process governs the future and how much the equity wrapper truly deserves to be discounted, remain the user's to judge.

Long horizons warrant added caution. Results for holding periods beyond approximately thirty years should be read as directional rather than precise. Across such spans several of the model's dynamics—the Bitcoin price trajectory, the structural-premium path, the credit-spread evolution, and the accumulation asymptote—grow increasingly sensitive to parameter choices, precisely where the framework extrapolates furthest beyond the data that inform it. The same horizon encompasses Bitcoin's transition from a block-subsidy-dominated to a fee-dominated security budget, multiple plausible monetary and interest-rate regimes, an evolving regulatory landscape, and successive changes in Strategy's capital structure, management, and strategy, none of which the framework conditions upon. The asymptotic outcomes the model produces at such horizons—in which a single entity approaches a substantial share of accessible supply—would themselves likely provoke second-order responses in network governance, regulation, and market structure that lie beyond the model's scope. Outputs past this horizon are therefore best interpreted as illustrative of qualitative tendencies rather than as quantitative projections.

12 Conclusion

Denominating value in Bitcoin turns the question of a treasury company's worth into a clean comparison against self-custody and exposes the premium as a function of three beliefs rather than a number to be argued in the abstract. The rational mNAV follows from one's Bitcoin outlook, one's trust in the corporate wrapper, and one's holding horizon, with the Bitcoin process the dominant lever, the wrapper hurdle a powerful and often neglected second, and the horizon an interaction term whose sign depends on conviction. Much of the public disagreement over Strategy's fair premium is, on this reading, a disagreement over these inputs conducted without making them explicit. The contribution here is a transparent engine that makes the premium the consequence of stated assumptions, so that debate can move to where it belongs: the assumptions themselves.

A Notation and symbols

This appendix collects the symbols used in the paper. Time-indexed quantities carry a subscript t, with 0 denoting the present (the calibration snapshot) and T the value at the holding horizon. Constants set as modeling defaults are listed with their base-case values in Table 4.

Symbol	Meaning
State: prices, holdings, shares	
B_t	USD Bitcoin spot price; B_0 today, B_T at the horizon.
H_t	Bitcoin held by the firm, in coins.
N_t	diluted common shares outstanding.
P_t	MSTR common share price.
b_t	Bitcoin backing per share, $b_t = H_t/N_t$.
Premium and valuation	
m_t	mNAV, the market premium: equity market capitalization P_tN_t over gross coin value H_tB_t.
$m_0^\star$	rational (fair) mNAV: the premium at which MSTR beats self-custody with probability one-half.
m^{eq}, m^{EV}, m^{net}	the three mNAV conventions of Section 2.1: equity-to-gross-coin (used here), enterprise-value, and net-asset.
m_{floor}	defensive common-ATM floor (issue to defend credit down to this premium).
m_{min}	structural floor on the premium function.
m_{acc}	premium above which the offensive (accretive) common ATM activates.
G	Bitcoin-denominated gross multiple of the position over the horizon.
v_T	exit value of one share in Bitcoin, $v_T = b_T\, m_T$.
V_0	fair MSTR price today, in dollars.
$\mathrm{med}(\cdot)$	cross-path median.
r_p	annual counterparty/wrapper hurdle.
τ	holding horizon, in years.
mNAV-convention snapshot (Section 2.1)	
P, N, H, B	snapshot share price, diluted shares, coins held, and spot price.
D	outstanding convertible principal.
L	aggregate preferred liquidation preference (snapshot); L_t below is its time-varying analogue.
π	(Section 2.1 only) the investor's assigned probability of a forced wind-down.
Short policy rate (SOFR) dynamics	
r_t	short policy rate (SOFR); r_0 today.
θ	long-run neutral rate.
κ	rate mean-reversion speed.

Symbol	Meaning
σ_r	rate volatility.
μ_J, σ_J	mean and volatility of rate jumps; λ_J the jump intensity (per year); Q_t the Poisson jump-counting process.
r^*	policy-rate cap threshold (the level above which the fiscal-dominance constraint pulls the rate down).
λ_{fd}	fiscal-dominance constraint strength (per-year downward pull on the excess above r^*).
W_t^r, W_t^B	Brownian drivers for the rate and for Bitcoin.
Bitcoin dynamics	
F_t	model fair value of Bitcoin (power-law line, or lag-corrected 200-week moving average).
n, A	power-law exponent (5.77) and coefficient, $F = A\,g^n$ in days g since genesis.
μ, σ	Bitcoin drift and volatility governing reversion toward the fair-value line.
σ_t	Bitcoin (residual) volatility; σ_0 initial, $\bar{\sigma}_t$ the trailing anchor, σ_{ref} a reference level.
ϕ_{dim}, ϕ_{gbm}	fair-value lag factors for the diminishing-returns and GBM models.
μ_B^J, σ_B^J	mean and volatility of the Bitcoin jump under the optional joint fiscal-stress shock.
Capital structure and waterfall	
S_t	preferred notional (aggregate liquidation preference).
D_t	convertible principal outstanding.
R_t	USD reserve; R_0 today.
L_t	senior liabilities ranking ahead of the common: preferred par, accrued arrears, and convertible principal.
a_t	amplification, $a_t = (S_t + D_t)/(H_t B_t)$; $a_{\text{tgt}}, a_{\max}, a_{\min}$ its target, ceiling, and floor.
η	accretive-issuance slope; $\bar{a}$ the market-depth (issuance-pace) cap.
ΔN	common shares issued in a step.
STRC pricing (required yield $\to$ coupon $\to$ price)	
P_t^{STRC}	STRC price; $\Pi = \$100$ its par.
q_t^{STRC}	required (market-clearing) STRC yield = SOFR + spread.
c_t	offered STRC coupon (state); ratchets toward $\min(q_t^{\text{STRC}}, c_t^{\max})$ in steps $\pm\Delta c$.
s_0, $\underline{s}$	inherent baseline spread over SOFR and the spread floor.
k_R, τ_R	reserve-runway adder coefficient and decay scale; $w_t = R_t/(\bar{q}_t S_t)$ the runway (yrs).
k_B, $\underline{a}$	senior-through-STRC coverage adder coefficient and its threshold amplification.

Symbol	Meaning
k_V, $\bar{\sigma}$	volatility adder coefficient and reference vol; d_t the distress add (pause 4% / forced BTC sale 2%).
$c_t^{\max}$, $\bar{c}_{\text{will}}$, ϕ	affordability-derived coupon cap, willingness ceiling, affordability horizon.
$\bar{p}$, A_{saw}	above-par price cap and ex-dividend sawtooth amplitude.
$\bar{q}^{\text{ceil}}$	hard rate ceiling above which STRC cannot clear even at par.
ε_t	standard normal shock.
Premium (mNAV) function	
ρ_t	net-backing ratio, $\rho_t = (H_t B_t - L_t + R_t)/(H_t B_t)$; $\widehat{\rho}_t = \text{clip}(\rho_t, 0, 1)$.
π_t	option/convexity premium; π_{floor} its durable floor and π_{cvx} its convexity coefficient.
π_{kick}	coefficient on the convex upside flywheel.
g_0	entry dislocation, $g_0 = m_0 - (\rho_0 + \pi_0)$; κ_π its decay speed.
Simulation	
n_{p}	number of Monte Carlo paths.
Δ	time step (one quarter).